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ARTICLE

Laboratory scale method for preparation of mixture modeled composite fuels for hybrid propulsion

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ABSTRACT

Fuel formulation is one of the chief strategies investigated in hybrid propulsion studies. In this context, polyethylenes-paraffins compositions have been suggested as a trade-off solution to attain both mechanical quality and ballistic performance. In parallel, mixed hybrids have already proven their capacity to enhance regression rates of thermoset polymers in mixture-modelled experiments. Ammonium perchlorate replacement by ammonium nitrate could turn mixed hybrids into environmentally benign materials. Meanwhile, the preparation of hybrid fuels based on paraffins and thermoplastic polymers seems to lack an application-oriented approach, capable to generate macroscopically homogeneous and symmetrical samples, compatible with both instrumental and ballistic measurements. Therefore, this work successfully outlines a laboratory scale preparation method developed to generate cylindrical composite fuel grains for hybrid propulsion, comprising low-density polyethylene (LDPE), macrocrystalline paraffin and ammonium nitrate mixture-modelled formulations, as well as applies thermogravimetry and differential scanning calorimetry to gain insight into their microstructural aspects, particularly their miscibility and its relation to the observed macroscopic behavior.

KEYWORDS

Hybrid propulsion; polyethylene; paraffin; ammonium nitrate; mixture modelling; preparation

Introduction

Solid fuel formulation is among the chief approaches aimed at improving hybrid propulsion feasibility, with composite fuels comprising paraffins and polymers receiving a great deal of attention due to the possibility of properly balancing regression rates, mechanical behavior and combustion efficiency (Ben-Basat and Gany 2016; Ishigaki and Nakagawa 2017; Tang et al. 2018). Accordingly, polymers of the polyethylene type seem promising alternatives not only because of their chemical compatibility with paraffins, but also as a result of their capacity to form a viscous melt layer, thus enabling, alongside

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with paraffin, the primary condition for entrainment occurrence and regression rate improvement (Karabeyoglu and Cantwell 2002; Kim, Moon, and Kim 2015; Kim et al. 2015). An important data resource on the microstructural interactions in polyethylenes-paraffins binary systems comes from phase change materials studies, whose results suggest the importance of thermogravimetry and differential scanning calorimetry in understanding their miscibility (Chen and Wolcott 2015, 2014; Hato and Luyt 2007; Krupa and Luyt 2001; Krupa, Miková, and Luyt 2007). Moreover, the paramount importance of miscibility has also been recognized in the solid propellants' field (Singh et al. 2019).

In parallel, mixed hybrids, characterized by the incorporation of a certain amount of a solid oxidizer to the fuel (Kuo and Chiaverini 2007), have been proved to enhance hydroxyl-terminated polybutadiene (HTPB) regression rates in hybrid rocket systems, with formulations defined according to the mixture modelling technique, in which the measured response is supposed to depend solely on the ingredients' proportions and not on the amount of the mixture (Cornell 1990; Frederick and Joshua Whitehead 2009). Vickland demonstrated the feasibility of an HTPB-ammonium perchlorate mixed hybrid motor operating in a tactical missile (Vickland 1967). The ever growing concerns about the impacts of the life cycle of propulsion systems on human health and on the environment (Brinck 2014) might motivate its replacement by ammonium nitrate, a green solid oxidizer (Chaturvedi and Dave 2013; Forsyth 2016; Jos and Mathew 2017). These observations suggest that a novel composite fuel comprising a polyethylene, a paraffin and ammonium nitrate may provide, under the hybrid propulsion standpoint, an interesting balance among their main isolated advantages, i.e., good mechanical quality, high regression rate and density increase/environmental friendliness, respectively.

In this context, the protocols adopted in the preparation of polyethylenes-paraffins composite samples and their microstructural interactions do not appear to have received in the open literature the same attention dedicated to the production of other fuels like pure paraffin samples (Andrianov et al. 2019; Piscitelli et al. 2015; Saccone et al. 2015; Stober et al. 2019) or thermoplastic ones such as acrylonitrile-butadiene-styrene copolymer (McKnight et al. 2015; Whitmore and Merkley 2019). Regarding pure paraffins (Saccone et al. 2015; Stober et al. 2019) and polyethylenes-paraffins composites (Boros and Konecný 2009; DeSain et al. 2009), centrifugal casting has been considered as a feasible technique due to its capacity to generate a simple geometry, characterized by a single and regular circular port, at acceptable costs.

Therefore, this work aimed at developing a laboratory scale preparation method for novel ternary composites comprising a polyethylene, a paraffin and ammonium nitrate, whose formulations stemmed from the mixture modelling technique (Cornell 1990), and at understanding their microstructural interactions from a thermal analysis standpoint. The preparation method was conceived as a two steps process, i.e., vigorous mixing at high temperature followed by centrifugal casting with natural cooling. Subsequently, thermal methods were used to provide both qualitative and quantitative insight into the microstructural behavior of the composites, allowing its interrelation to macroscopical aspects, i.e., homogeneity and the existence of serious structural defects such as major cracks and voids.

Materials and Methods

Considering the importance of viscosity to the enhancement of the fuel's regression rate and to the adequate processability of the ternary composites, it was selected a low-density polyethylene grade with a high Melt Flow Index (MFI), supplied by Braskem and known as PB608, hereinafter designated as LDPE. According to the supplier, it exhibited a 28 g/10 min MFI (190°C/ 2.16 kg). Macrocrystalline paraffin was chosen bearing in mind the reduction of the melt layer viscosities. Hence, the Petrobras variety 140/145-2 was selected, with a melting temperature of 61.4°C, from this point on labeled simply as paraffin. Finally, analytical grade ammonium nitrate was supplied by Merck and presented a nominal purity of 99.8%. Prior to use, the material was grounded, sifted and dried. The $297<\Phi<590$ μm fraction was selected for the preparation of the samples.

The preparation of the composite samples was conducted using an aluminum mold, approximately 5.0 cm in height and 7.0 cm in external diameter, provided with threadable lids. The aluminum mold also had bottom and top polytetrafluorethylene (PTFE) stopper lids to prevent leakages. The aluminum mold was conceived to operate both as a container for the mixing step and as a cast mold. The heating in the mixing step was accomplished by means of a glycerin bath assembled on a heating plate provided with a temperature controller. A mechanical stirrer provided with a digital tachometer was used. The centrifugal casting was conducted with the aluminum mold at the horizontal position, by means of another mechanical stirrer. The development of the preparation method will be discussed in the next section.

The compositional consistency in the radial axis was assessed through Thermogravimetric (TGA) and Differential Scanning Calorimetry (DSC) analyses, carried out with samples extracted from three consecutive radial portions of 1 mm thick discs, previously cut on a lathe from the middle longitudinal portion of each of the composite grains. Macroscopic homogeneity and the existence of major structural failures were assessed by visual inspection of the composite grains themselves and of their 1 mm thick discs.

TGA and DSC measurements were carried out on a Netzsch STA449F3 simultaneous thermal analyzer, previously calibrated with KNO_3, Ag_2SO_4, K_2CrO_4 and $BaCO_3$ salts, selected due to the use of platinum crucibles and whose solid-solid transitions match the temperature range of interest (NETZSCH-

Gerätebau GmbH 2012). The experimental conditions were set as follows: 20 mL/min N_2 protective atmosphere, 80 mL/min synthetic air purge atmosphere, 10°C/min heating-cooling rate, round-shaped samples, roughly 1.0 mm thick and 5.0 mm in diameter. The temperature program comprised an initial dynamic step (heating) from 30°C to 140°C, followed by a first isothermal step at 140°C for 10 min, by an intermediate dynamic step (cooling) from 140°C to 30°C, by a second isothermal step at 30°C for 50 min and by a final dynamic step (heating) from 30°C to 600°C. The samples' mass ranged from 10 to 25 mg. The crucibles weren't capped by lids because of the simultaneous interest in the mass changes and in the heat fluxes developed under a partially oxidative atmosphere (Haines et al. 2002). Definition of the experimental regions was conducted under the mixture modelling principles, specifically the extreme vertices technique (Cornell 1990), using MINITAB19 software.

Results and Discussion

A. Development of the Composites' Preparation Method: First Experimental Region

The process was expected to be set up with common laboratory equipment, to be safe and capable to provide cylindrical fuel grains containing a single circular port, yielding both macroscopic homogeneity and dimensional regularity. The first attribute was understood as the result of both compositional consistency in terms of the system polymer-paraffin, and nearly uniform dispersion of the solid oxidizer in the radial axis. Dimensional regularity meant the absence of serious structural defects such as cracks and major voids, as well as the attainment of a symmetrically shaped port. At this point, some exploratory experiments were conducted, comprising pure macrocrystalline paraffin and its 50/50 wt% binary compositions with LDPE and ammonium nitrate, as well as two ternary formulations, as can be seen in Table 1.

From these screening experiments, the rotation regime during mixing was divided in two phases, the first one at 50 rpm to promote the initial mixing under a considerable thermomechanical stress. Then, the second mixing phase

Table 1. Formulations of the composites prepared in the screening step.

		Original components proportions (wt%)		
Order	Code	A (LDPE)	B (Paraffin)	C (NH_4NO_3)
1	2 V	50.00	50.00	0.00
2	3 V	50.00	25.00	25.00
3	5 V	25.00	25.00	50.00
4	1 V	00.00	100.00	00.00
5	4 V	00.00	50.00	50.00

at 100 rpm for roughly 15 min was set to complete homogenization. The rotation speed (400 rpm) during centrifugal casting (120 min) should represent a convergence between the formation of a regular circular port and an approximately uniform dispersion of the oxidizer. This preliminary work showed that the assessed formulations had a very poor homogeneity level. The only exceptions in terms of homogeneity were pure paraffin and its 50/50 wt% composition with ammonium nitrate, labeled 1 V and 4 V in Table 1, which, on the other hand, featured composite grains with a quite poor mechanical quality. Another important point was the exceptionally high hygroscopicity of the 50 wt% NH_4NO_3 formulations, i.e., 4 V and 5 V in Table 1, which may hinder the fuel's ignitability (Jos and Mathew 2017) and negatively affect its thermal decomposition (Manelis et al. 2003).

These elements led to the definition of the first experimental region, conceived to validate the operational settings. The formulations in this region were determined using the mixture modelling principles, specifically the extreme vertices technique (Cornell 1990). The quantitative limits adopted for each of the raw materials were as follows: LDPE ranged from 10.00 to 30.00 wt%, paraffin ranged from 30.00 to 90.00 wt% and ammonium nitrate ranged from 0.00 to 40.00 wt%. These intervals kept paraffin as the major component, thus favoring the entrainment mechanism by the minimization of the molten fuel's dynamic viscosity (Karabeyoglu, Altman, and Cantwell 2002; Karabeyoglu and Cantwell 2002), while LDPE main role would be to provide the mechanical backbone. Table 2 outlines the formulations generated by MINITAB19 based on these limits. No replicates were included due to the still exploratory role of this stage. The formulations in Table 2 comprise four vertices (denoted by "V"), four middle edge points (denoted by "C(1)") and an overall centroid (denoted by "C(2)"). This domain includes polyethylenes-paraffins formulations commonly assessed in hybrid propulsion studies (Kim, Moon, and Kim 2015; Kim et al. 2015). The preparation of the composite samples was conducted under the operational settings discussed above. The corresponding grains are shown in Figure 1, with the numbers matching the preparation sequence defined in Table 2.

Table 2. Formulations of the composites prepared in the first experimental region.

			Original components proportions (wt%)		
Order	Code	Identification	A (LDPE)	B (Paraffin)	C (NH_4NO_3)
1	2 V	New_2 V_01	30.00	70.00	0.00
2	5 C(1)	New_5 C(1)_00	20.00	80.00	0.00
3	9 C(2)	New_9 C(2)_00	20.00	60.00	20.00
4	3V	New_3V_00	30.00	30.00	40.00
5	6 C(1)	New_6 C(1)_00	10.00	70.00	20.00
6	7 C(1)	New_7 C(1)_00	30.00	50.00	20.00
7	8 C(1)	New_8 C(1)_00	20.00	40.00	40.00
8	1V	New_1V_00	10.00	90.00	0.00
9	4V	New_4V_00	10.00	50.00	40.00

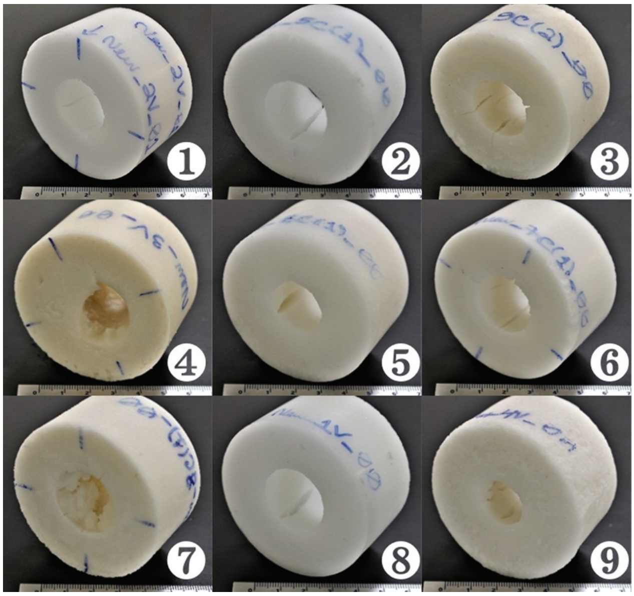

Figure 1. Composite grains produced in the first experimental region.

Upon analysis, it was observed that all the composite grains featured major structural defects such as deep cracks and fissures distributed randomly across the port, as can be seen in Figure 1. This occurred independently of the presence of NH_4NO_3. The macroscopic homogeneity assessment was conducted by visual inspection of the composite grains themselves and of their 1 mm thick discs. It revealed clearly the heterogeneous dispersion of the solid oxidizer and confirmed the existence of deep cracks and major fissures, almost evenly distributed. Structural defects have also been observed in a commonly studied formulation, comprising 10 wt% polyethylenes/90 wt% paraffins (Kim et al. 2015; Pal, Kumar, and Li 2019), represented in this study by composite New_1 V_00. Further attempts to improve homogeneity through adjustments in the centrifugal casting speed did not bring the expected results. The structural defects were not eliminated either. Moreover, despite the acceptable processability level attained with most of the formulations in this first experimental region, the upper limit for NH_4NO_3 (i.e., 40 wt%) proved itself unpractical, for it hampered the formation of a circular port symmetrically shaped.

These experimental findings can be associated to a new figure of merit, i.e., the ratio between the amounts of paraffin and LDPE in each formulation, designated "B/A", for these letters were attributed to the materials in Table 2. In the first experimental region, B/A ranged from 1.0 (New_3 V_00) to 9.0 (New_1 V_00). Among the oxidizer containing formulations, the ones with the higher B/A ratios, i.e., New_6 C(1)_00 (B/A = 7), New_4 V_00 (B/A = 5) and New_9 C(2)_00 (B/A = 3) yielded the worst macroscopic homogeneities.

This was not observed, for instance, with composite New_3 V_00 (B/A = 1), which exhibited a homogeneous oxidizer dispersion. On the other hand, the hygroscopicity of the high-content oxidizer formulations was still excessive. Moreover, the screening preparations conducted prior to this first experimental region had suggested that the absence of structural defects could be attained with compositions having a low B/A ratio. This set of observations thus motivated a dislocation of the experimental region, keeping the preparation protocol defined at this stage unaltered. Specifically, LDPE proportion was shifted to the 30.00– 50.00 wt% range while the paraffin amount was changed to the 40.00– 60.00 wt% interval, in order to improve the dimensional regularity and to get a better radial dispersion pattern for NH_4NO_3. The oxidizer proportion was narrowed to the 0.00– 20.00 wt% range, to facilitate processability and to reduce hygroscopicity. In addition, reflecting the apparent importance of B/A ratio both to homogeneity and symmetry, the linear multicomponent constraint $1.1 \leq B/A \leq 1.5$ was applied (Cornell 1990).

As for the preparation method so far developed, variations in the main operational factors such as rotation speed during centrifugal casting were not able to circumvent the described defects. This suggested a direct relation between these nonconformities and the formulation range, which should be dislocated in order to attain macroscopic homogeneity and dimensional regularity. Besides, the preparation protocol displayed the fundamental features pursued from the beginning, i.e., simplicity, safety and scalability, while generating circular single port cylindrical grains. Thus, it was considered ready for the purposes of producing grains analyzable both in lab scale hybrid motors as well as by instrumental techniques such as TGA and DSC. It can be outlined as follows: (1) Raw materials weighing and mold assembly; (2) LDPE and paraffin pre-mixing in the mold, under 120 rpm, at ambient temperature; (3) Mold immersion in the glycerin bath at 150°C, melting and mixing at 50 rpm, until the bath temperature returns to the initial setting; (4) Stirring at 100 rpm, with the bath temperature at 150°C, for 15–25 minutes; (5) Slow addition of $297<\Phi<590$ μm NH_4NO_3, under 100 rpm stirring and with the bath temperature at 150°C; (6) Homogenization for 15 min at 100 rpm, bath temperature kept constant at 150°C; (7) Removal of the stirrer, upper stopper and lid assembly, horizontal centrifugal casting at 400 rpm for 120 min, followed by demolding.

B. Development of the Composites' Preparation Method: Second Experimental Region

The limits adopted for each of the raw materials stemmed from the abovementioned restrictions. Accordingly, LDPE ranged from 32.00 to 47.62 wt %, paraffin ranged from 41.91 to 60.00 wt% and ammonium nitrate ranged from 0.00 to 20.00 wt%. Table 3 outlines the formulations generated by

Table 3. Formulations of the composites prepared in the second experimental region.

			Original components proportions (wt%)		
Order	Code	Identification	A (LDPE)	B (Paraffin)	C (NH_4NO_3)
1	9 C(2)	Nouv_9 C(2)_00	39.43	50.57	10.00
2	7 C(1)	Nouv_7 C(1)_00	42.86	47.14	10.00
3	8 C(1)	Nouv_8 C(1)_00	36.00	54.00	10.00
4	2V	Nouv_2V_00	47.62	52.38	0.00
5	4V	Nouv_4V_00	32.00	48.00	20.00
6	1V	Nouv_1V_00	40.00	60.00	0.00
7	9 C(2)	Nouv_9 C(2)_01	39.43	50.57	10.00
8	6 C(1)	Nouv_6 C(1)_00	35.05	44.95	20.00
9	5 C(1)	Nouv_5 C(1)_00	43.81	56.19	0.00
10	9 C(2)	Nouv_9 C(2)_02	39.43	50.57	10.00
11	3V	Nouv_3V_00	38.10	41.90	20.00

MINITAB19, organized in the random order of preparation. As before, formulations were defined under mixture modelling principles, in order to allow further model fitting and analysis of parameters of interest, such as dynamic viscosity, melting enthalpy and thermal expansion (Cornell 1990). Two centroid replicates were included to allow repeatability assessment. The preparation method previously outlined was successfully applied to these formulations, with a good processability level being observed throughout all the batches. The composite grains so produced are shown in Figure 2. The numbers match the random preparation order defined in Table 3.

Comparing the grains prepared in the first (Figure 1) and second (Figure 2) experimental regions, a considerable improvement was attained under the simultaneous perspective of macroscopic homogeneity and dimensional regularity. This can be assessed by the direct comparison of centroids, which reflect the average behavior of their respective experimental domains (Cornell 1990). In fact, composite grain New_9 C(2)_00 (number 3 in Figure 1) exhibits the typical appearance observed in the first experimental region, with major cracks and fissures across the port and a poor homogeneity, while composite grain Nouv_9 C(2)_00 (number 1 in Figure 2) shows a circular port free from noticeable structural defects and a much better radial NH_4NO_3 dispersion. As for dimensional regularity, the only exceptions were the composites Nouv_8 C (1)_00 and Nouv_4 V_00, which developed some minor cracks across the port. These grains were the only ones, among those containing solid oxidizer, whose B/A ratio was 1.5, i.e., the upper limit for this parameter in the second experimental region.

TGA and DSC analyses allowed some qualitative and quantitative insights into the homogeneity and symmetry levels attained by the preparation protocol. Figure 3 shows the TGA (a) and the DSC (b) traces of the samples collected at the outer, middle and inner positions of a given radial segment of the disc from centroid composite Nouv_9 C(2)_00. All the thermal curves refer to the last dynamic heating step (30°C – 600°C).

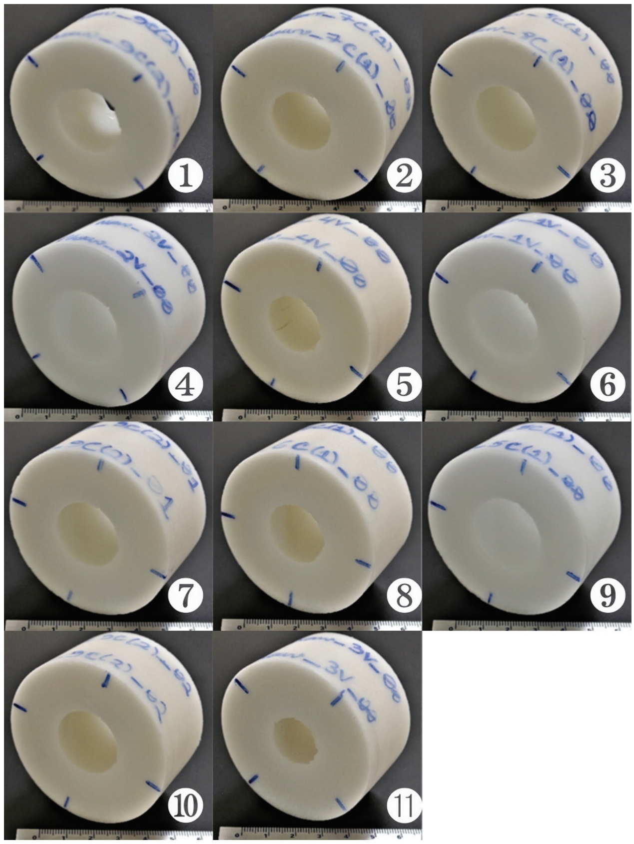

Figure 2. Composite grains produced in the second experimental region.

Both TGA and DSC traces suggest reasonably close profiles for melting and thermal decomposition stages. In Figure 3 (a), the trace for the outer position exhibits a higher mass loss from the start of the final dynamic step (circa 30°C), probably due to a higher concentration of NH_4NO_3 on this region, whose moisture was lost during the first dynamic heating up to 140°C. Accordingly, in Figure 3 (b) the noticeable difference in the melting stage (up to circa 200° C) lies on the peaks associated with a solid-solid transition of NH_4NO_3 (128° C) and with its melting (169°C) on the outer position trace. As for thermal decomposition, both traces suggest its occurrence in three consecutive stages, among which the second one is the less distinct, possibly due to the overlapping of decomposition reactions under the selected experimental conditions. TGA traces indicate that the mass losses for the outer, middle and inner positions in the 30°C – 600°C range were 99.82%, 99.40% and 99.74%, respectively, amounting to an average mass loss of 99.65% and a relative standard deviation (Skoog et al. 2004) of 0.22%. These results point out that, under a partially oxidative atmosphere, the attained NH_4NO_3 radial dispersion

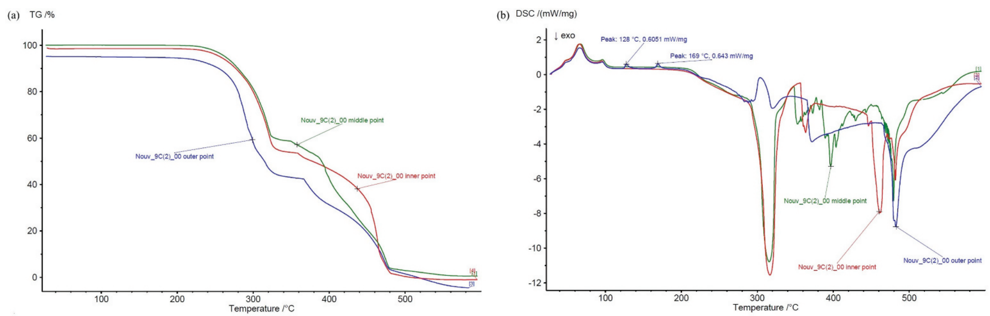

Figure 3. (a) TGA and (b) DSC traces of samples collected from a given radial segment of composite disc Nouv_9 C(2)_00.

Table 4. Some quantitative data for samples collected from a given radial segment of composite disc Nouv_9 C(2)_00.

Sample	Onset temperature (°C)	Offset temperature (°C)	Melting enthalpy (J/g)	Melting peak temperature (°C)
Nouv_9 C(2)_00 outer position	276.8	478.8	202.2	66.0
Nouv_9 C(2)_00 middle position	277.2	485.9	230.4	66.7
Nouv_9 C(2)_00 inner position	283.1	477.1	218.7	65.4

produced a uniform mass loss behavior along the burning direction, i.e., across the radial axis. Table 4 presents some other data extracted from Figure 3.

Applying Dixon criterion (Skoog et al. 2004) for the detection of outliers to each data subset in Table 4 reveals that none of the extreme results (suspects) differs significantly from the respective data subset, at the 95% confidence level. This confirms that NH_4NO_3 radial dispersion does not seem to affect the thermal behavior along the burning direction. Furthermore, these quantitative findings corroborate the choice of the middle radial position as a representative one for sampling the discs for thermal analyses. As for the repeatability assessment of the preparation method, Figure 4 presents the TGA traces of the centroid replicates and the raw materials, obtained, as before, in the final dynamic heating step.

From the qualitative standpoint, the traces for the composite's replicates exhibit an intermediary behavior as compared to the raw materials, as expected from other studies (Cardoso et al. 2017; Kim, Moon, and Kim 2015). This seems particularly true for the beginning of thermal decomposition and for the existence of three subsequent decomposition stages in all three

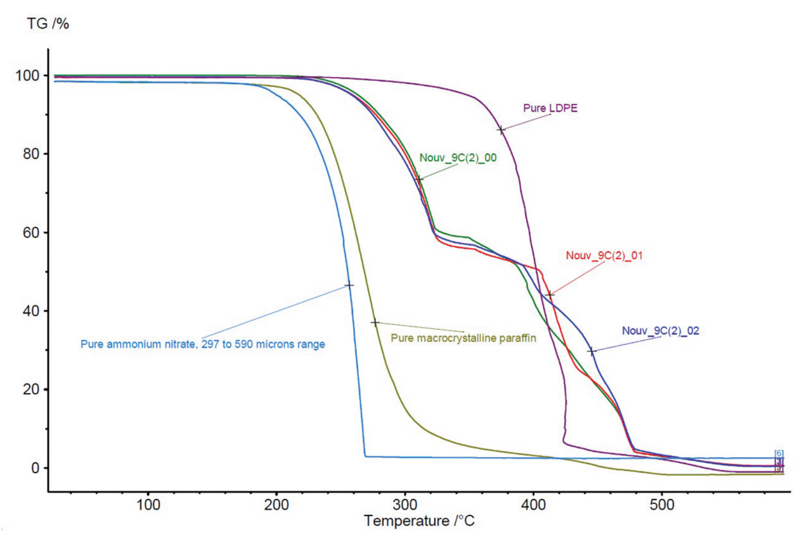

Figure 4. TGA traces of the centroid replicates obtained in the second experimental region.

replicates. The good repeatability of the centroid replicates results, suggested by the proximity of their TGA traces, can be quantitatively assessed as well. Application of the Dixon criterion to the onset temperatures of the centroid replicates, i.e. 277.2°C for Nouv_9 C(2)_00, 300.6°C for Nouv_9 C(2)_01 and 259.0°C for Nouv_9 C(2)_02, yielded no rejection of the extreme (suspect) values, at the 95% confidence level, thereby showing that from both qualitative and quantitative standpoints, the overall variability resulting from the preparation protocol seems quite satisfactory.

The dimensional assessment of the composite grains at 20°C showed that length ranged from 3.52 to 3.60 cm, external diameter ranged from 6.50 to 6.55 cm and internal diameter ranged from 2.24 to 3.05 cm, while the mass ranged from 79.39 to 95.32 g. Data for a pure macrocrystalline paraffin grain obtained according to the described preparation method were taken as well, yielding a mass of 81.91 g, a length of 3.64 cm, while external and internal diameters amounted to 6.60 and 2.76 cm, respectively. From these data it was possible to calculate the volumetric contraction of the grains. In the second experimental domain it lied between 6.99% and 18.01%, with an average contraction of 10.84%. Moreover, the contraction of the pure macrocrystalline paraffin grain yielded a 9.90% figure, while literature points out a 13.0 to 25.0% range for pure paraffins under static conditions (Saccone et al. 2015; Stober et al. 2019). This difference possibly stemmed from the turbulent condition associated to centrifugal casting, in the sense that the slow accommodation of paraffin molecules is not feasible under vigorous rotation. In short, these observations suggest that cooling pure paraffin at rest enhances its crystalline character and hence augments its volumetric contraction, the contrary being true for centrifugal casting.

The expressive improvement in terms of composite grains' quality resulting from experimental region dislocation raises the matter of the miscibility degree attained between LDPE and paraffin and its effects on dimensional regularity. DSC allowed this assessment (Haines 1995; Winters 2012). Accordingly, the crystallization profiles for binary LDPE-paraffin composites ranging from 10.00 wt% to 47.62 wt% LDPE content, as well as for pure paraffin and pure LDPE, were obtained in the cooling step from 140°C to 30°C. The DSC traces are presented in Figure 5. This data set comprises both the first and second experimental regions and the results discussed from now on refer to a single measurement for each composite.

As expected, all the binary composites yielded intermediary traces between their pure raw materials, thus confirming the existence of different levels of miscibility, i.e., microstructural interactions between LDPE and paraffin (Haines 1995; Winters 2012). DSC traces allowed the calculation of the temperature gap between the two major peaks identifiable for the composites, each of them corresponding to LDPE and paraffin phases and to their maximum crystallization rates (Louanate et al. 2020). The results are organized in

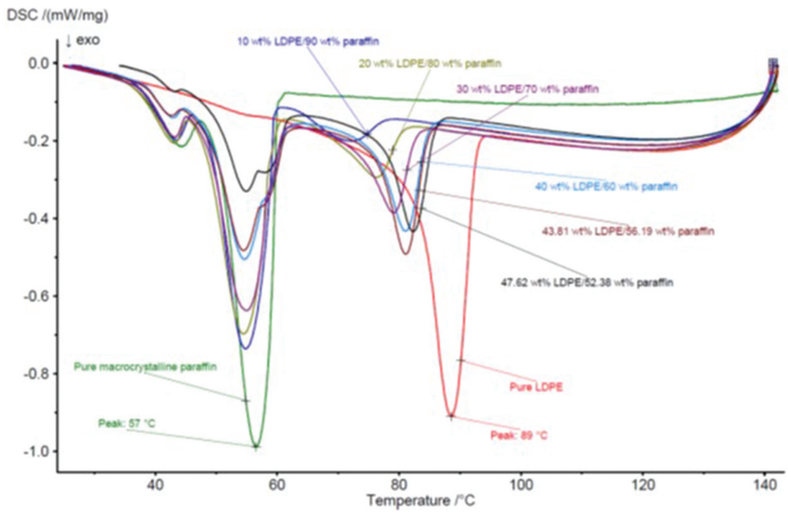

Figure 5. DSC traces of the binary LDPE-paraffin composites during the cooling step.

Figure 6 (a). They also provided crystallization (or solidification) enthalpies, i.e., the heat released during cooling of the composites as well as of pure raw materials. They are presented in decreasing order in Figure 6 (b), where "0.00% LDPE" stands for pure paraffin.

The temperature gap values are all lower than 32°C, the figure corresponding to pure LDPE and paraffin traces. This corroborates the occurrence of a set of miscibility levels between LDPE and paraffin. Moreover, the growing tendency of this figure with the LDPE content suggests that the accommodation of paraffin molecules in a branched polymer like LDPE becomes ever more difficult with its increasing proportions, i.e., their miscibility decreases. These observations led to the conclusion that in the second experimental region (40.00 wt% to 47.62 wt% LDPE) there is a partial miscibility between LDPE and paraffin.

In Figure 6 (b) three distinct groups of crystallization enthalpies can be seen. The first of them corresponds to the pure materials, with the lowest absolute values, i.e., −213.5 J/g and −290.2 J/g. Next, there is a group of enthalpies in the −320.0 J/g range, corresponding to the second experimental region. Finally, there exists the −380.0 J/g range, the highest absolute solidification enthalpies, exactly attributable to the first experimental region. The absolute data indicate the existence of a considerable synergistic interaction between LDPE and paraffin in cooling, under the overall experimental conditions adopted, away from the additive rule (Kim, Moon, and Kim 2015). It is also valid to point out that this interaction reached its maximum in the first experimental region, characterized by a very poor dimensional regularity (see Figure 1). In addition, a joint assessment of the data in Figure 6 suggests that a higher miscibility between LDPE and paraffin was in fact

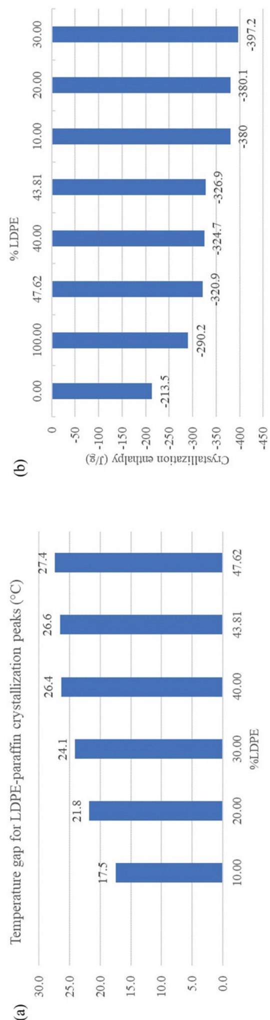

Figure 6. (a) Temperature gap for LDPE-paraffin crystallization peaks and (b) crystallization enthalpies for the LDPE-paraffin composites.

attained in the first experimental region. This resulted in the release of higher amounts of energy during solidification, as compared to the second experimental region, which yielded a lower miscibility level and a corresponding lower energy release during crystallization. From the standpoint of the first experimental region, these observations mean that, under the centrifugal casting conditions adopted, a higher amount of energy had to be dissipated in a shorter time interval associated to the lower temperature gap found in this range. The overall result in terms of dimensional regularity appears to be the existence of serious structural defects, as shown in Figure 1. Finally, in view of the attained solid oxidizer range, it is important to emphasize that the ternary formulations herein discussed are not intended for use as solid propellants by themselves, but as fuels in hybrid rocket motors.

Conclusions

The laboratory scale batch method for the preparation of composite fuels for hybrid propulsion comprising LDPE, paraffin and ammonium nitrate was successfully developed, yielding a protocol consisting of two major steps, i.e., vigorous mixing at high temperature and natural cooling centrifugal casting. This method allowed the repetitive production of single circular port cylindrical grains, with a volumetric contraction in the 6.99% – 18.01% range, prone to be used both in static firings and in instrumental analysis such as TGA and DSC, thus establishing an initial commitment between ballistic (i.e., regression rate) and analytical assessments. Once the nature of the raw materials was defined and the outline of the batch preparation method was set, the expected interdependence between the structural quality of the composite grains and their formulation range was confirmed, being necessary to dislocate this range in order to produce acceptable grains. TGA and DSC results suggested that the partial miscibility level between LDPE and paraffin attained in the adjusted experimental region, yielding a crystallization enthalpy in the −320.0 J/g range, was enough to provide a good compositional uniformity in the radial axis of the grains and a crystallization behavior compatible with the inexistence of major cracks. Under the overall experimental conditions adopted, the volumetric contraction of pure paraffin, i.e. 9.90%, was found to be lower than literature data, as well as crystallization enthalpy results suggested a synergistic interaction between LDPE and macrocrystalline paraffin, away from the additive rule. The devised preparation method and its macroscopic and microstructural experimental findings are expected to be useful in hybrid propulsion studies, regarding particularly the proper definition of novel LDPE-macrocrystalline paraffin-NH_4NO_3 composite fuels' formulations.

Declaration Of Interest

The authors declare that there is no conflict of interest associated to this work.

Nomenclature

A = LDPE label in mixture modelling
B = Macrocrystalline paraffin label in mixture modelling
C= Ammonium nitrate label in mixture modelling
V = Vertices in mixture modelling
C = Centers in mixture modelling

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